

Emmanuel Despi

Software Engineer · Alabang, Muntinlupa City · 09056248875 · emmanuelrobisodespi@gmail.com

EXPERIENCE

Senior Software Engineer · Kivomotion (Outsourced PH)

May 2025 – Apr 2026

- Owned end-to-end development of a responsive, high-performance Flutter companion app across iOS, Android, and Web for a multi-sport wearable band; transformed raw sensor data into training metrics including velocity, acceleration, stress, and sport-specific KPIs.
- Designed data ingestion and ETL/ELT pipelines for high-volume, real-time wearable sensor data; validated and transformed raw telemetry before loading analytics data into DynamoDB.
- Engineered a serverless AWS backend using Amplify, API Gateway, Lambda, DynamoDB, S3, CloudFront, and Cognito; selected NoSQL and serverless architecture for scalable, cost-efficient delivery.
- Established code review, unit/integration testing, and CI/CD practices; managed technical debt through MVP iterations and maintained production stability.
- Mentored engineers on Flutter, AWS, and platform architecture; established development standards and documentation.
- Partnered with product owners, designers, and stakeholders to translate requirements into technical sprints and communicate constraints and trade-offs.
- Implemented JWT/OAuth authentication, IAM policies, secure health/fitness data handling, CloudWatch logging, and performance monitoring.

Robotics Software Engineer · Dyson

Dec 2023 – Apr 2025

- Developed and maintained a C++/CMake multi-subsystem platform targeting a dual-MCU architecture for a robot vacuum product.
- Designed and led hardware pytest infrastructure; established Python/XML testing standards and implemented unit, integration, and hardware testing with gtest; owned Bitbucket CI/CD pipelines.
- Partnered with product owners and scrum masters to communicate requirements, impediments, blockers, and hardware-integration needs.
- Owned production features including cloud synchronization, logging, OTA updates, diagnostics, and troubleshooting; drove reliability and bug-fix improvements.
- Configured Linux hardware-test environments and used AWS S3/EC2, Docker, Jira, Confluence, and Bitbucket for deployment and collaboration.

Embedded Software Engineer · Thales

Jul 2022 – Nov 2023

- Developed and maintained a C++ Key Management System for smart-card OS; implemented encryption/decryption logic for debit and credit card transactions.
- Optimized client-specific card encryption/decryption modules for performance and reliability.
- Built rigorous automated testing in Python, XML, and Shell scripts against client requirements and team standards.
- Collaborated in a 4-person team across Waterfall sprints and with a French counterpart team for integration and production releases.

Software Developer · TESDA

Sept 2021 – Jun 2022

- Maintained and developed 5+ government systems covering scholarship handling, student/teacher data encryption, partner-school management, procurement, and budget allocation.
- Owned full-stack delivery across ASP.NET MVC, Razor, MSSQL, and AWS; implemented scholarship allocation, student employment tracking, filtering, reporting, and data-governance features.
- Delivered on-call departmental requests, including custom tax and budget computation modules, and owned AWS component deployment.
- Trained TESDA branches and partner schools during BSRS rollout; provided technical support and documentation and managed Jira sprint delivery.
- Tech stack: C# ASP.NET MVC, Razor, MSSQL, AWS.

CIE Operator · FactSet

Jul 2020 – Mar 2021

- Monitored and troubleshooted production workflows processing real-time stock-market data; performed root-cause analysis on workflow failures.
- Documented investigations and remediation recommendations; maintained real-time tickets through resolution.

Intern Software Engineer · Nokia

Dec 2019 – Mar 2020

- Worked under tech-lead supervision and participated in Scrum ceremonies, sprint planning, retrospectives, and technical decision-making.
 - Implemented a C++/Python diagnostic system for Nokia 5G networks in Linux, including a self-evaluation module for detecting and reporting connection issues.
 - Designed and built a peer-evaluation web application for quarterly employee reviews as a capstone project.
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SELF-EMPLOYMENT

Software Engineer · Self-employed

2020 – Present

- Designed, architected, and delivered custom enterprise applications including POS, ordering, and business-management systems from requirements through deployment and client support.
 - Built responsive applications with Flutter, React, and React Native across mobile and web; selected technologies based on client requirements and constraints.
 - Designed serverless AWS backends with Amplify, API Gateway, Lambda, DynamoDB, S3, CloudFront, and Cognito; established scalable NoSQL architectures and infrastructure practices.
 - Architected REST APIs with RBAC/IAM and JWT/OAuth authentication for multi-tenant and single-tenant applications.
 - Applied Provider, Riverpod, Bloc, and React state-management patterns; maintained code quality through testing, reviews, and documentation.
 - Managed client requirements, technical trade-offs, delivery, and post-launch support.
 - Set up CI/CD, Docker, Linux scripting, monitoring, and AWS scaling/cost optimization.
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TECHNICAL SKILLS

Languages — C, C++, C#, Python, Dart, SQL, Shell Script, JavaScript, TypeScript, HTML, CSS

Frontend — Flutter, React.js, React Native, SwiftUI, Tailwind CSS, Material UI, ASP.NET MVC, .NET Core

Backend & Data — Node.js, Firebase, REST APIs, ETL/ELT, Real-time Processing, Event Streaming

Databases — MSSQL, MySQL, MongoDB, DynamoDB (NoSQL), Relational & Columnar Models

Cloud & DevOps — AWS (S3, Lambda, API Gateway, Amplify, EC2, CloudWatch), Docker, Kubernetes, Linux, CI/CD (Bitbucket, GitHub Actions)

State Management — Provider, Riverpod, Bloc; Component-based Architecture; Performance Optimization

Security — RBAC/IAM, JWT/OAuth, Encryption, Secure Data Handling, Data Governance

Tools & Practices — Git, Atlassian (Jira, Confluence, Bitbucket), Postman, Figma, Agile/Scrum, Waterfall

Hardware — Microcontrollers, Sensors, Smart Card Systems, 5G Network Systems

EDUCATION

B.S. Computer Engineering · Graduated May 2020

Far Eastern University – Institute of Technology

Specialization: MCU-based Prototypes · Thesis: Bionic Arm with Haptic Feedback

CERTIFICATIONS

- Career Service Professional — Civil Service Commission, Philippines